PRODUCT SCOPE · INPUT TO SPEC 0.3

Minmed Digital Name Cards

A digital name card for every Minmed staff member, managed centrally. Admins create and change every card. Staff sign in only to see their card, add a name or alias to their link and download their QR code. Anyone with a card's link can open it, but nobody can find it on Google.

Based on Technical Specification draft 0.2 (15/10/2026) and your decisions on 16/10/2026. Names, numbers, codes and QR codes in the examples are made up. Updated 16/10/2026.

CONFIRMED you decided it PROPOSED my recommendation, waiting for your OK (section 11) TEST try on a test app before the build

1

How it works

1 Admin adds the card
In the admin console, one at a time or from a spreadsheet.

2 Staff member gets an email
Their link and QR code, an Add to Google Wallet button, and steps for Apple Wallet.

3 Staff member can add a name to the link
Optional. They sign in with their Minmed Google account and type a name or alias. They get a new link and a new card email; the old link stops working.

4 Staff member shares it
Sends the link on WhatsApp or email, or shows the QR code from Wallet.

5 Visitor opens the link
Sees the card, then saves the contact, emails, calls or WhatsApps. Unusual traffic gets a CAPTCHA.

6 Details change
The staff member replies to the card email. An admin edits the card, and the same link shows the new details.

7 Staff member leaves
HR tells an admin. On the last day the card is deactivated and soft-deleted, and the staff member can't sign in. The link shows “This card is no longer active.”

Figure 1. The life of a card, from creation to deactivation.

2

What's confirmed

On 16/10/2026 you approved the proposals in this document. Open items are in section 11.

Decision	Source
PRODUCT	
One central place to manage every staff card. Staff can't change their card details, so there's no approval process. The only thing they can change is the name or alias in their link.	You, 16/10
Cards are public to anyone with the link, but can't be found through Google.	You, 16/10
Each link ends in a random code of 22 letters and numbers, so nobody can guess it: `me.minmed.sg/card/qB8Rh9sTxKuaSF3YkGmD2w`. Staff can put a name or alias in front of the code: `me.minmed.sg/card/Alex-Tan/qB8Rh9sTxKuaSF3YkGmD2w`.	You, 16/10; industry standards
Staff sign in with their Minmed Google account to a My card page at `me.minmed.sg/my-card`. There they can only read their card details, set or change the name or alias in their link, and download their QR code. They keep their link on their phone, or in Apple Wallet or Google Wallet.	You, 16/10
Built on Base44. Phone-first public card. Admins and staff sign in with company Google accounts.	Spec 0.2 §1.1; you, 16/10
THE CARD	
Fields: name, designation, Minmed, work email, optional mobile and photo, optional biography, languages and LinkedIn. Company information is a link to minmed.sg; no social links.	Spec 0.2 §3.1; you, 16/10
Actions: Save contact and Email, plus Call and WhatsApp when a mobile is published. No visitor app.	Spec 0.2 §1.1; you, 16/10
Every card shows the Minmed logo. A Super admin sets it once in Settings; it's the same on every card and can't be changed for a single card. The card forms and My card show it as locked.	You, 16/10
On a computer, the card also shows a QR code to move it to a phone.	You, 16/10
No “Active staff profile ✓ Checked” line and no How to verify panel.	You, 16/10
Sharing the link on WhatsApp shows a preview of the card. **AT RISK** Base44 can't make a per-card preview as is. See section 10 and open item 1.	Spec 0.2 §4.4
WALLET	
Minmed issues a Google Wallet pass. iPhone staff add their QR code to Apple Wallet themselves with iOS 27's Create a Pass. No Apple Developer Program, fee or D-U-N-S number.	You, 16/10
NAME OR ALIAS IN THE LINK	
Optional. Letters, numbers and hyphens, 2 to 30 characters.	You, 16/10
Until an alias is set, the plain link without a name works.	You, 16/10
Setting or changing the alias gives a new link. The old link, printed QR codes and Apple Wallet passes made from it show “No card found”. The Google Wallet pass updates itself, and a new card email goes out automatically.	You, 16/10
Staff can change their alias as often as they like. Each change ends the previous link, and My card asks them to confirm before saving.	You, 16/10
Clearing an alias, by the staff member or an admin, brings back the plain link without a name.	You, 16/10
Before saving, staff can preview their card at the new link. Nothing changes until they save: the new link doesn't work yet and the old one keeps working.	You, 16/10
Admins can clear an alias that's offensive or copies someone else's name. Every alias change is logged and emailed to manage@minmed.sg.	You, 16/10

Decision	Source
ADMINS	
Two roles: Super admin and Admin.	You, 16/10
Only a Super admin can reactivate (restore) a card, give it a new link code or delete it.	You, 16/10
Cards can be imported from a spreadsheet at launch.	You, 16/10
Admins see view counts. No monthly report emails.	You, 16/10
Every admin change is logged and emailed to manage@minmed.sg.	Spec 0.2 §1.1
LEAVERS AND RECORDS	
HR tells an admin as part of offboarding. On the person's last day their card is deactivated and soft-deleted, and they can no longer sign in.	You, 16/10
Soft-deleted cards are kept but hidden. A Super admin can restore one or delete it permanently.	You, 16/10 (PDPA)
Soft-deleted cards are listed under a Deactivated tab on the Cards page, which only Super admins see.	You, 16/10
BOT PROTECTION	
One IP address making more than 3 card opens or contact downloads within 1 minute must pass a CAPTCHA.	You, 16/10
After passing, that browser isn't asked again for 30 minutes.	You, 16/10
A busy card is checked: more than 20 opens of one card in 10 minutes means everyone opening it gets the CAPTCHA for the next 30 minutes. This replaces “every 5th visitor”.	You, 16/10; industry standards
A pass ends early after 20 opens in 5 minutes.	You, 16/10; industry standards
For the first 2 weeks the rules only record. Then they switch on.	You, 16/10; industry standards

3

Features

Everything in the first version. Anything not listed here isn't being built.

Public card

`me.minmed.sg`

- Card page: photo, name, designation, contacts and About
- Save contact, Email, Call and WhatsApp
- QR code when opened on a computer
- WhatsApp link preview
- No longer active, not found and temporary error pages
- CAPTCHA check against bots
- Hidden from search engines

Staff

`me.minmed.sg/my-card`

- Minmed Google sign-in, own card only
- My card: read-only details, a name or alias for the link, a preview before saving, QR code download
- Card email with link, QR code and Wallet options
- Google Wallet pass that updates itself
- Apple Wallet pass they make themselves
- Ask for changes by replying to the card email

Admin console

`me.minmed.sg/admin`

- Google sign-in, admin list only
- Cards: list, search, add, import, edit
- Card details: link, alias (clear it if needed), QR code, card image, views, history
- Deactivate (soft delete); Super admin also restores, re-codes and deletes permanently
- Activity log
- Admins and Settings, including the company logo (Super admin)
- Email to manage@minmed.sg on every change, alias changes included

4

Users

User	Signs in?	Who they are	What they do
Visitor	No	Anyone who has a card's link or scans its QR code.	Views the card, saves the contact, gets in touch. May be asked to pass a CAPTCHA.
Staff member (card holder)	Yes, company Google account, to their own My card page only	Minmed staff with a card.	Gets their card by email, keeps it in Wallet or on their phone, shares it. Can add a name or alias to their link and download their QR code. Asks an admin for other changes.
Admin	Yes, company Google account	For example HR, added by a Super admin.	Adds, edits, sends and deactivates cards.
Super admin	Yes, company Google account	One or two owners, for example manage@minmed.sg's owner and IT.	Everything an Admin does, plus admins, settings, and reactivating, re-coding or deleting cards.

manage@minmed.sg receives admin notifications automatically. It's a mailbox, not a user.

5

Roles and permissions

✓ allowed · — not allowed · **Own** their own card only · **Auto** the system does it. Every rule is checked on the server, not only by hiding buttons.

Action	Visitor	Staff member	Admin	Super admin
PUBLIC CARD				
Open a card from its link or QR code	✓	✓	✓	✓
Save contact, email, call, WhatsApp	✓	✓	✓	✓
Forward the link to someone else	✓	✓	✓	✓
Find a card without its link (search, browse, guess)	—	—	—	—
GETTING AND KEEPING A CARD				
Receive the card email (link, QR code, Wallet options)	—	**Own**	—	—
Add the card to Google Wallet or Apple Wallet	—	**Own**	—	—
Sign in to My card and see their own card details	—	**Own**	—	—
Set or change the name or alias in their own link	—	**Own**	—	—
Preview their card at a new link before saving	—	**Own**	—	—
Clear their own alias, going back to the plain link	—	**Own**	—	—
Download their own QR code	—	**Own**	—	—
Change their own card details	—	—	—	—
Ask for a change by replying to the card email	—	**Own**	—	—
ADMIN CONSOLE				
Sign in to the admin console	—	—	✓	✓
See and search all cards	—	—	✓	✓
Add a card	—	—	✓	✓
Import cards from a spreadsheet	—	—	✓	✓
Edit a card's details and photo, and choose whether the mobile is shown	—	—	✓	✓
Clear a staff member's alias	—	—	✓	✓
Send or re-send the card email	—	—	✓	✓
Download a card's QR code and card image	—	—	✓	✓
Deactivate a card (soft delete)	—	—	✓	✓
Restore a deactivated card	—	—	—	✓
Issue a new link code (the old link stops working)	—	—	—	✓
Delete a card permanently	—	—	—	✓
See view counts	—	—	✓	✓
See the activity log	—	—	✓	✓

Action	Visitor	Staff member	Admin	Super admin
Add or remove admins and change roles	—	—	—	✓
Change the company logo shown on every card (in Settings)	—	—	—	✓
Use a different logo on one card	—	—	—	—
Change the brand colours, company details, notifications and Google Wallet settings	—	—	—	✓
AUTOMATIC				
Record every admin action and alias change in the activity log		**Auto**		
Email manage@minmed.sg when a card is added, edited, given a new alias, deactivated, restored, re-coded or deleted		**Auto**		
Send a new card email and update the Google Wallet pass when an alias changes		**Auto**		
Update the Google Wallet pass when its card changes, and expire it when the card is deactivated		**Auto**		
Ask for a CAPTCHA when a bot-protection rule is met (section 8)		**Auto**		

6

Site map and navigation

Four parts: the public card, the staff's My card page, the admin console, and the emails and passes that connect them. Select a page to see its mockup in section 7.

Public

`me.minmed.sg`

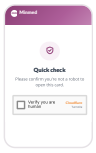
No sign-in · hidden from search engines · no menu



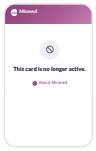
Card page
`/card/{code}`
or /card/{alias}/{code}



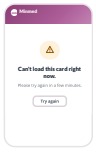
On a computer
adds a QR code



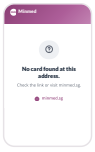
CAPTCHA check
when a rule in section 8 is met



No longer active
after deactivation



Temporary error
card can't be checked



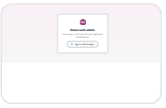
Card not found
any wrong address

- **Files behind the same code**
  `/functions/…?code={code}`
  contact file, card image, Google Wallet link; served by backend functions
- **Home**
  `/`
  redirects to minmed.sg; no staff list

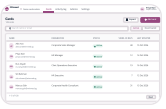
Admin console

`me.minmed.sg/admin`

Company Google sign-in · admin list only



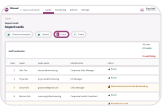
Sign in
`/admin`



Cards
`/admin/cards`



Add card
`/admin/cards/new`



Import cards
`/admin/cards/import`

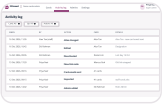


Card details
`/admin/cards/{id}`

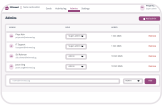


Edit card
`/admin/cards/{id}/edit`

Confirm dialogs
deactivate, clear alias, new link code, delete



Activity log
`/admin/activity`



Admins
`/admin/admins`
Super admin only



Settings
`/admin/settings`
Super admin only

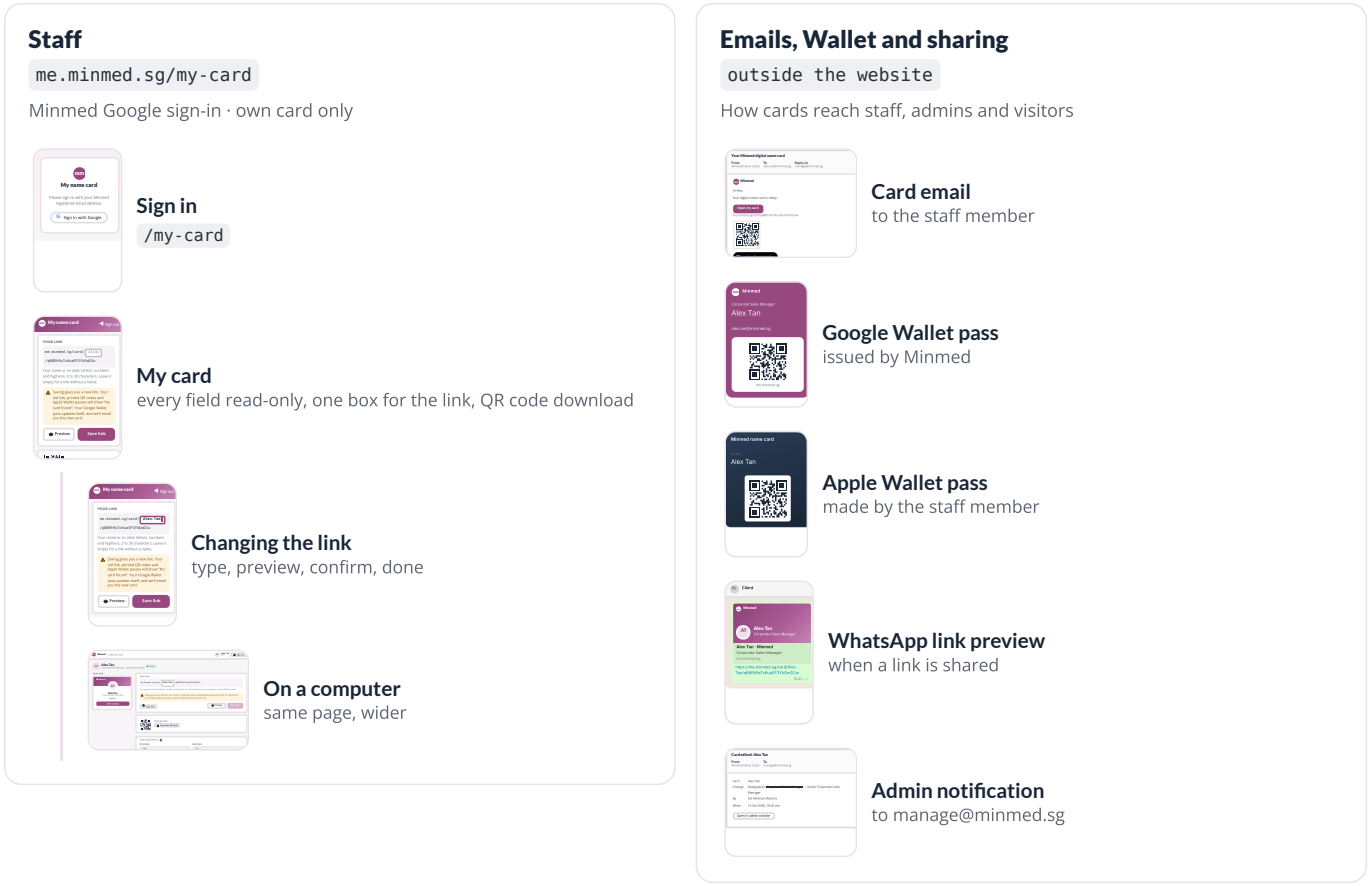


Figure 2. Site map. Each entry is a page or message, with its address.

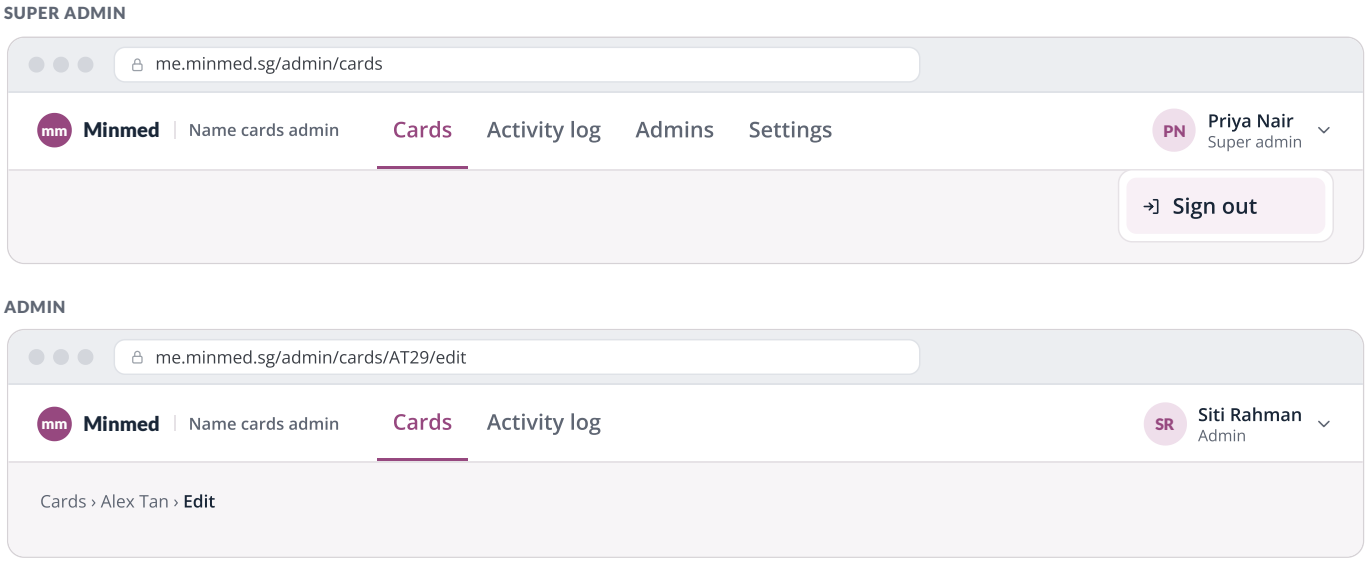


Figure 3. Admin console navigation.
Admins see Cards and Activity log; Super admins also see Admins and Settings. The account menu holds Sign out. Inside a card, the breadcrumb leads back to Cards.

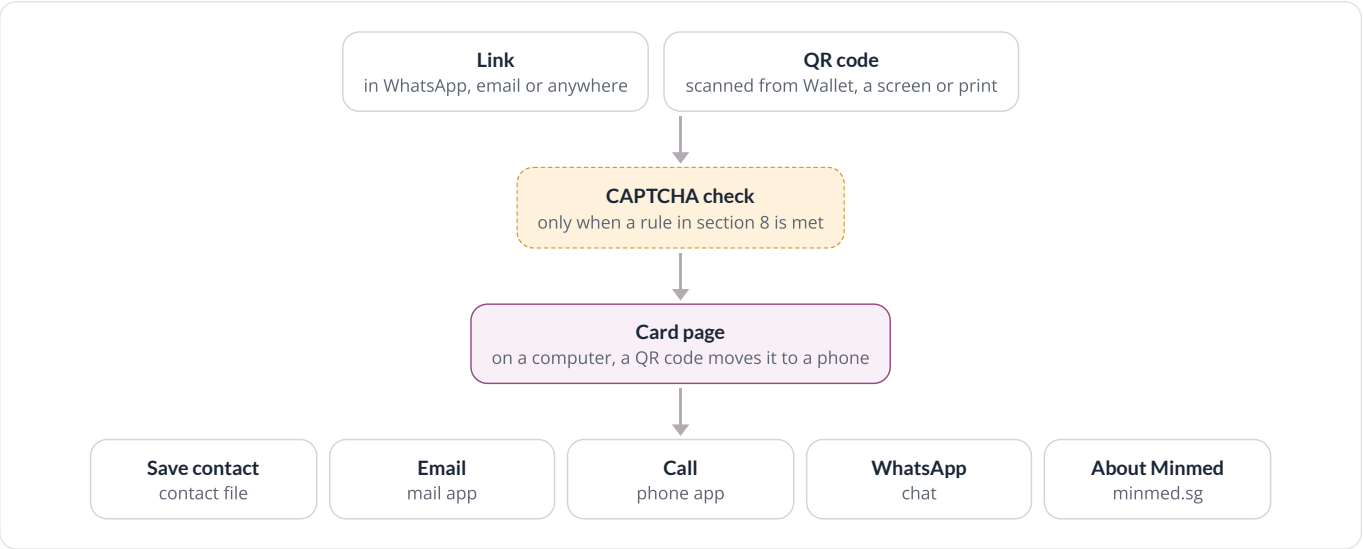


Figure 4. How a visitor moves through the public card.
There's no menu and no way to reach another person's card. Call and WhatsApp appear only when a mobile is published.

7

Pages

High-fidelity mockups of every page and message. You approved the wording on 16/10/2026, except the preview screen's (open item 2). The “mm” logo stands in for the real one, and the QR codes can't be scanned.

Public card

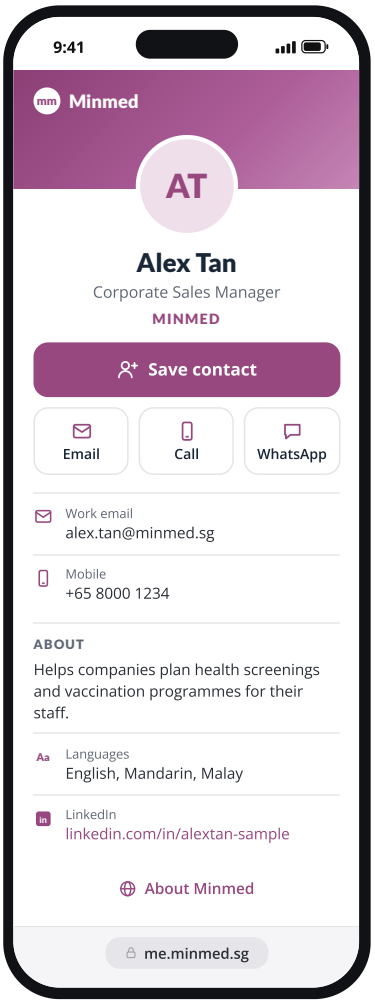


Figure 5. Card page.
`me.minmed.sg/card/Alex-Tan/qB8Rh9sTxKuaSF3YkGmD2w`, or `me.minmed.sg/card/qB8Rh9sTxKuaSF3YkGmD2w` until a name or alias is added · anyone with the link. Call and WhatsApp only with a published mobile; About only when filled in.

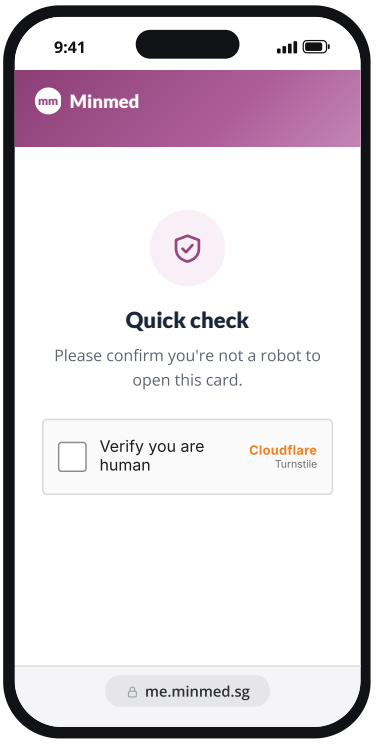


Figure 6. CAPTCHA check.
Same address, shown instead of the card when a rule in section 8 is met. Most people pass without clicking; this is what a flagged visitor sees.

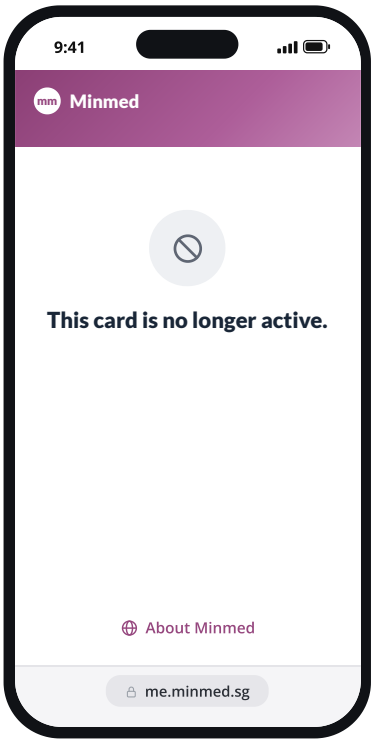


Figure 7. No longer active.
Same address after deactivation. No details, photo, Save contact or contact buttons.

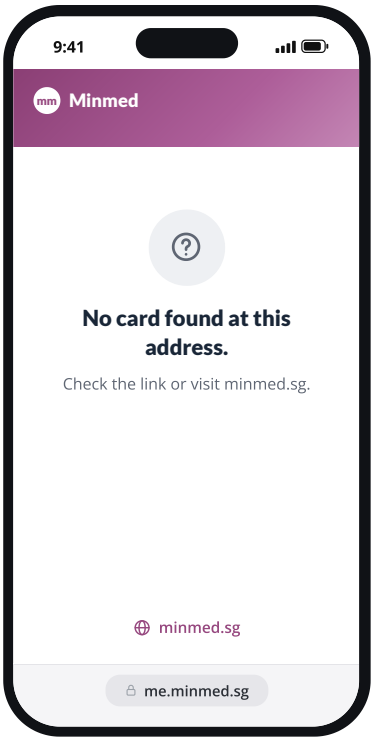


Figure 8. Card not found.
Any wrong code, old alias, deleted card or mistyped link. The same answer every time, so nobody learns which codes exist.

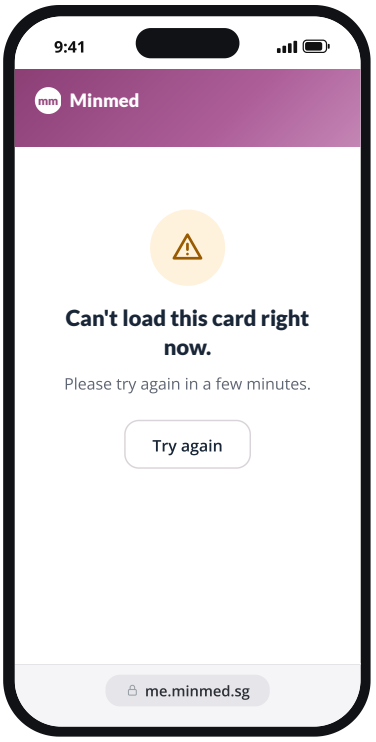


Figure 9. Temporary error.
The card can't be checked, for example during an outage or when Base44 credits run out. Never shows old details.

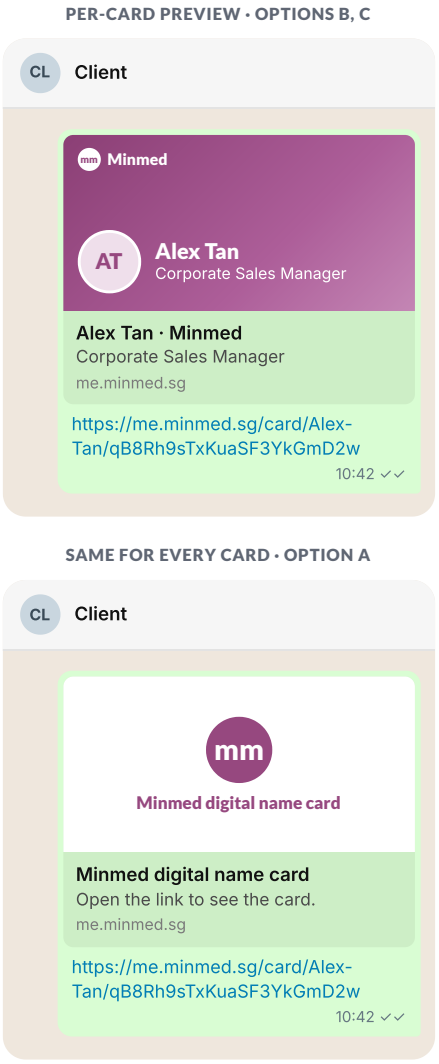


Figure 10. WhatsApp link preview. TEST
Which one depends on the WhatsApp test (open item 1).

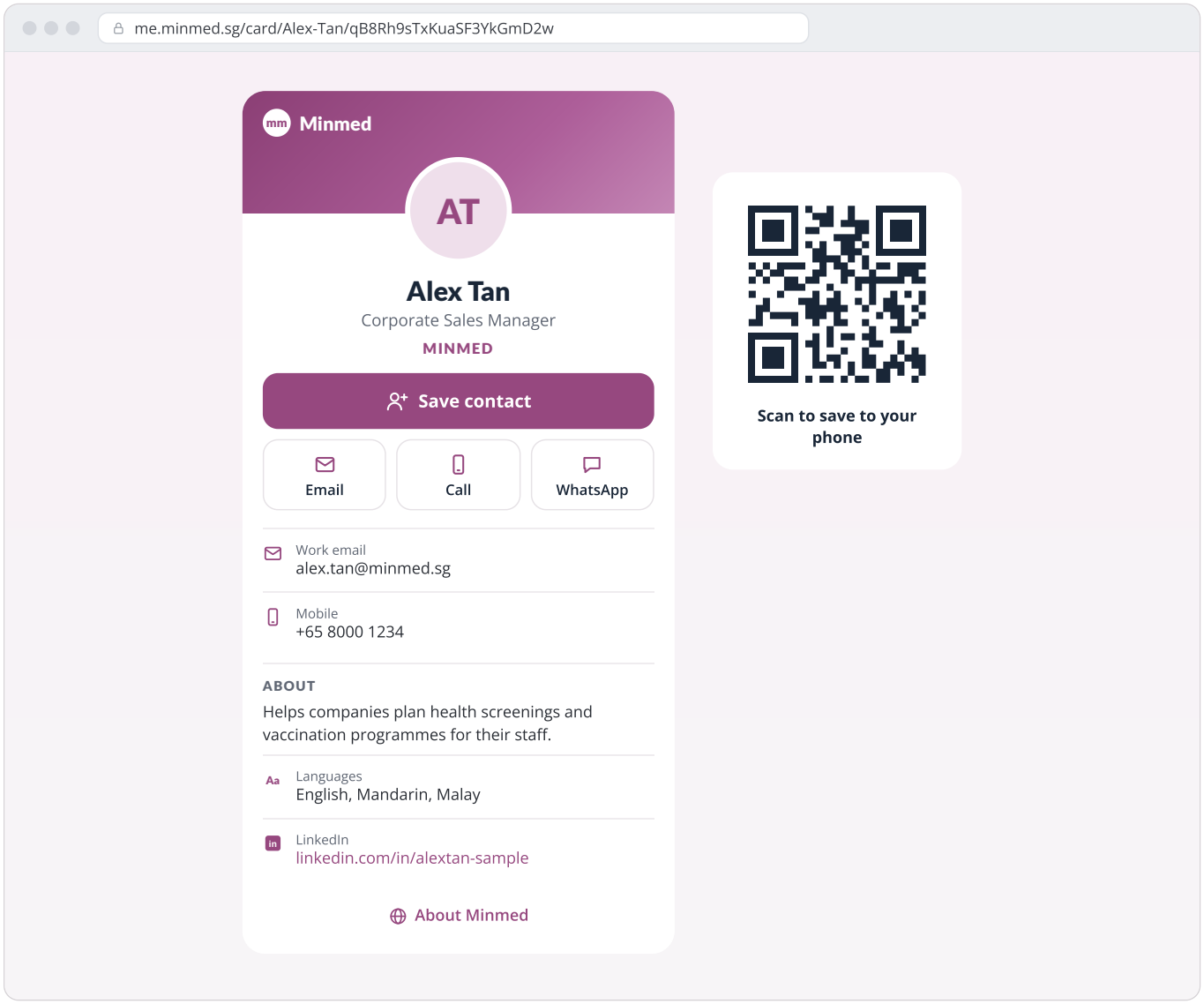


Figure 11. Card page on a computer.
The same card in a narrow column, plus a QR code to move it to a phone.

Staff

Your Minmed digital name card

From Minmed Name Cards
To alex.tan@minmed.sg
Reply-to manage@minmed.sg

Minmed

Hi Alex,

Your digital name card is ready.

Open my card

me.minmed.sg/card/qB8Rh9sTxKuaSF3YkGmD2w

Add to Google Wallet

iPhone (iOS 27)

1. Take a screenshot of the QR code.
2. In Wallet, tap + then Create a Pass.
3. Choose the screenshot.

No Apple Intelligence? Choose Create Pass Manually and scan the QR code from a computer screen.

Want your name in your link? Sign in at me.minmed.sg/my-card.

Need another change? Reply to this email.

Figure 12. Card email to the staff member.
Sent when an admin chooses Save and send card email or Send card email, and again after every alias change. Replies go to manage@minmed.sg.



Figure 13. Google Wallet pass.
Issued by Minmed. Updates when the card changes and shows as expired when it's deactivated.

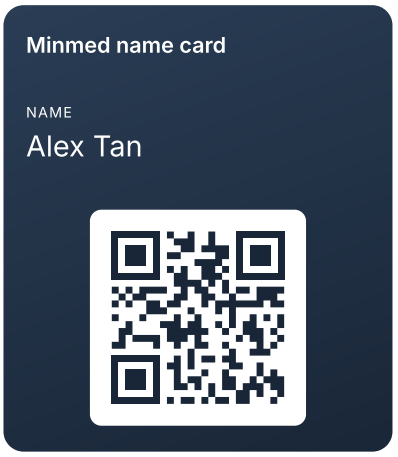


Figure 14. Apple Wallet pass, made by the staff member.
Made with iOS 27's Create a Pass, so its look may vary. Minmed can't update or remove it, but its QR code always opens the live card.

My card (staff view)

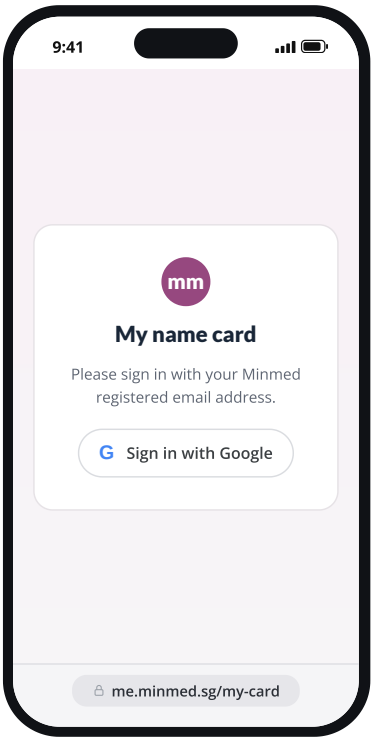


Figure 15. My card sign-in.
`me.minmed.sg/my-card`. A Minmed account without an active card sees "There's no name card for this account."

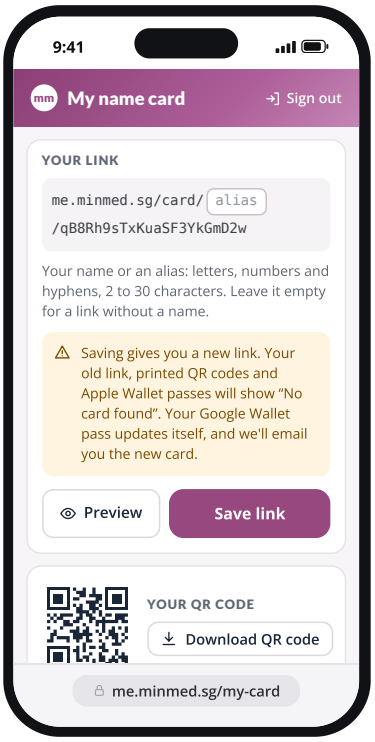


Figure 16. My card.
Top of the page after signing in. The only box staff can type in is the name or alias in their link. Preview shows the card at the new link before saving, and they can download their QR code. Below that, every field is read-only, including the company logo (whole page in Figure 18).

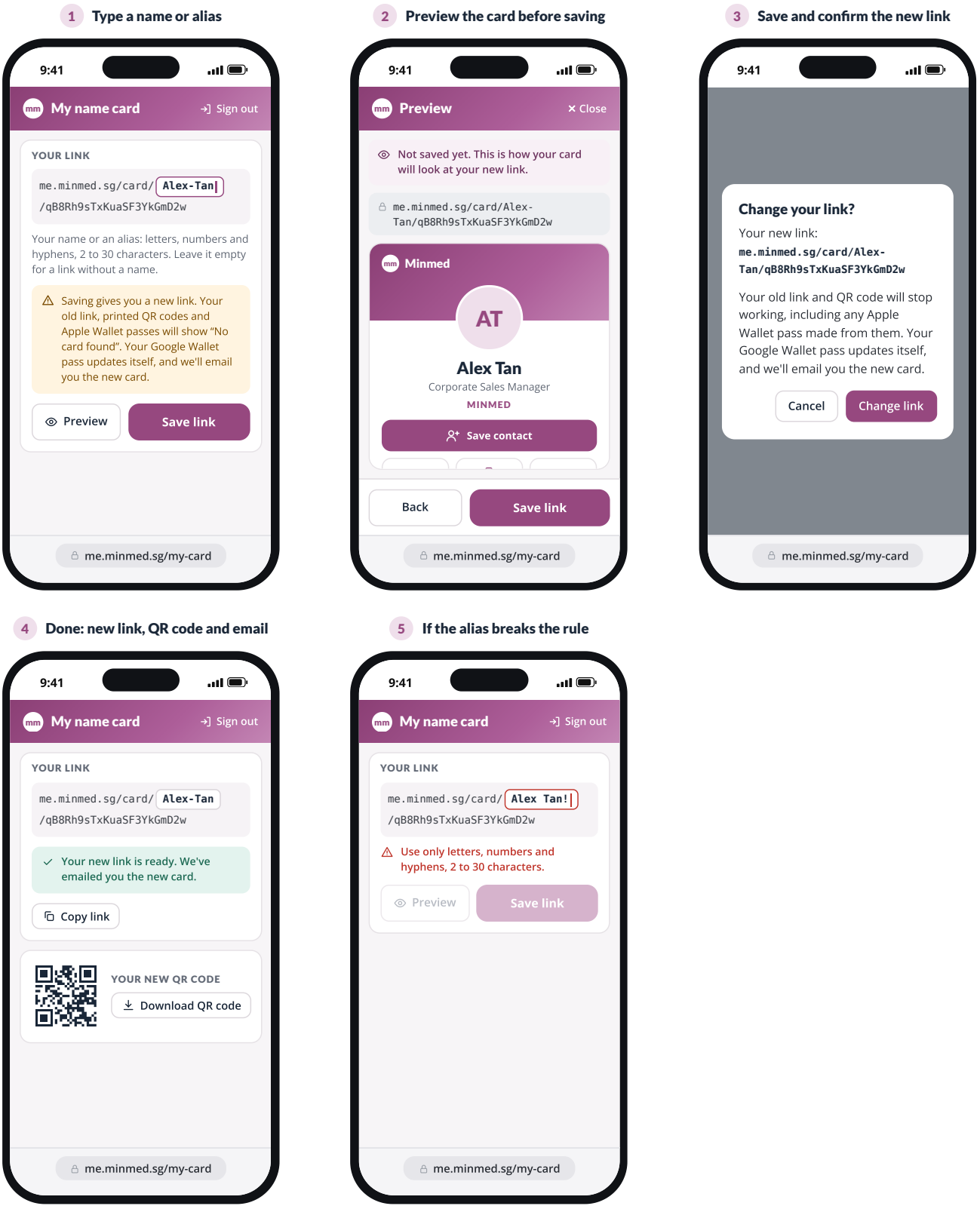


Figure 17. How a staff member changes their link.

1. They type a name or alias in the box. 2. Preview shows their card at the new link; nothing is saved yet. 3. They save and confirm. 4. The new link works at once and a new card email goes out; the old link shows "No card found". 5. An alias that breaks the rule can't be previewed or saved.

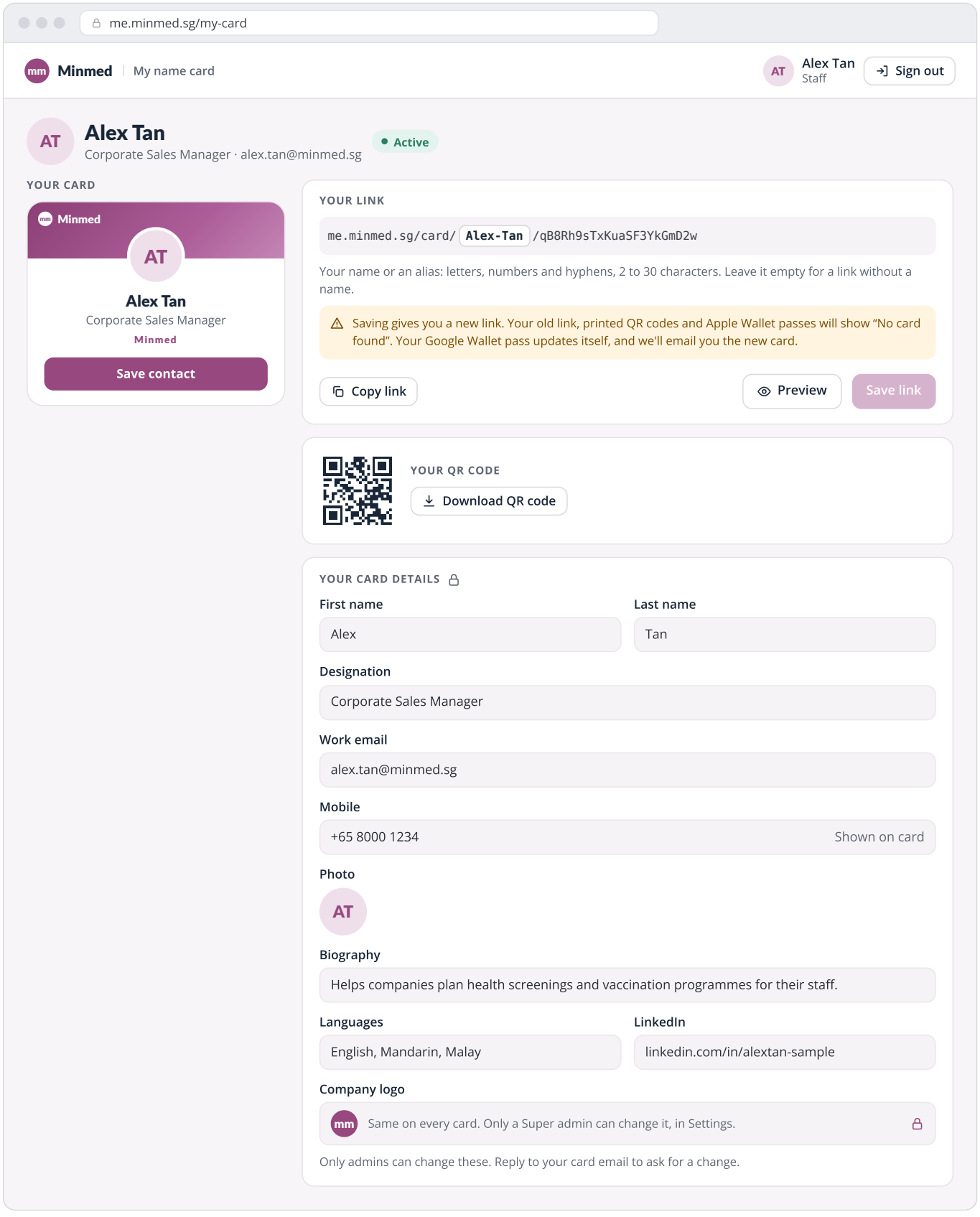


Figure 18. My card on a computer.
The same page, laid out like the admin card details page: the card on the left, then the link box, the QR code and every field read-only.

Admin console

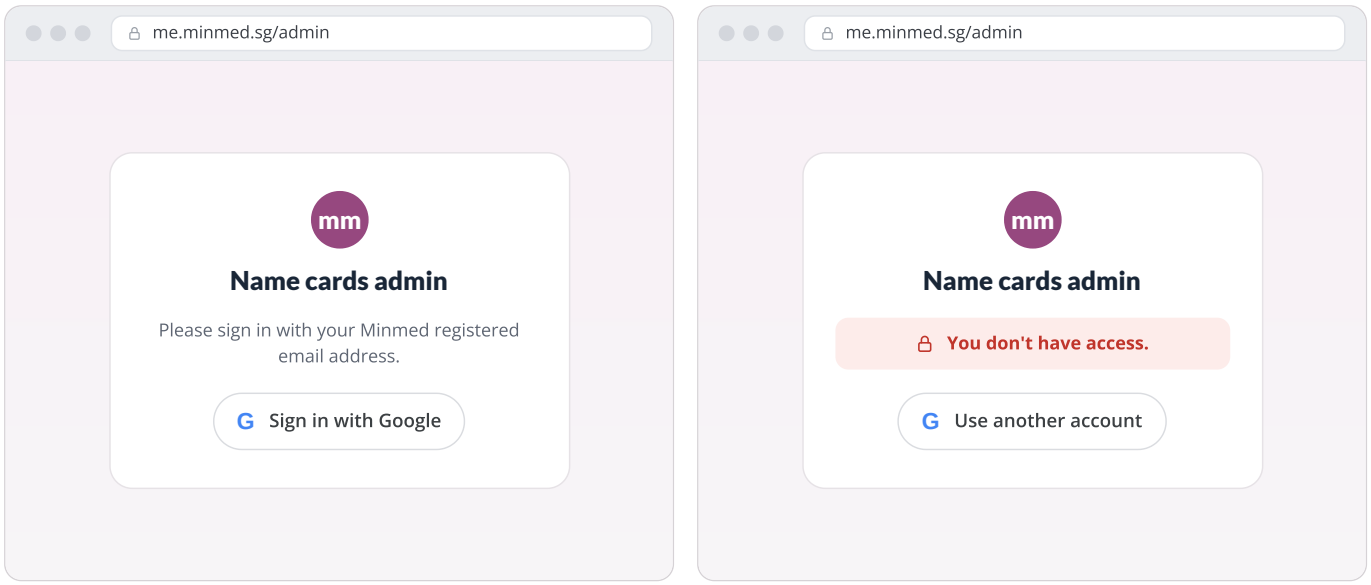


Figure 19. Admin sign-in.
Left: `me.minmed.sg/admin`. Right: a company account that isn't on the admin list.

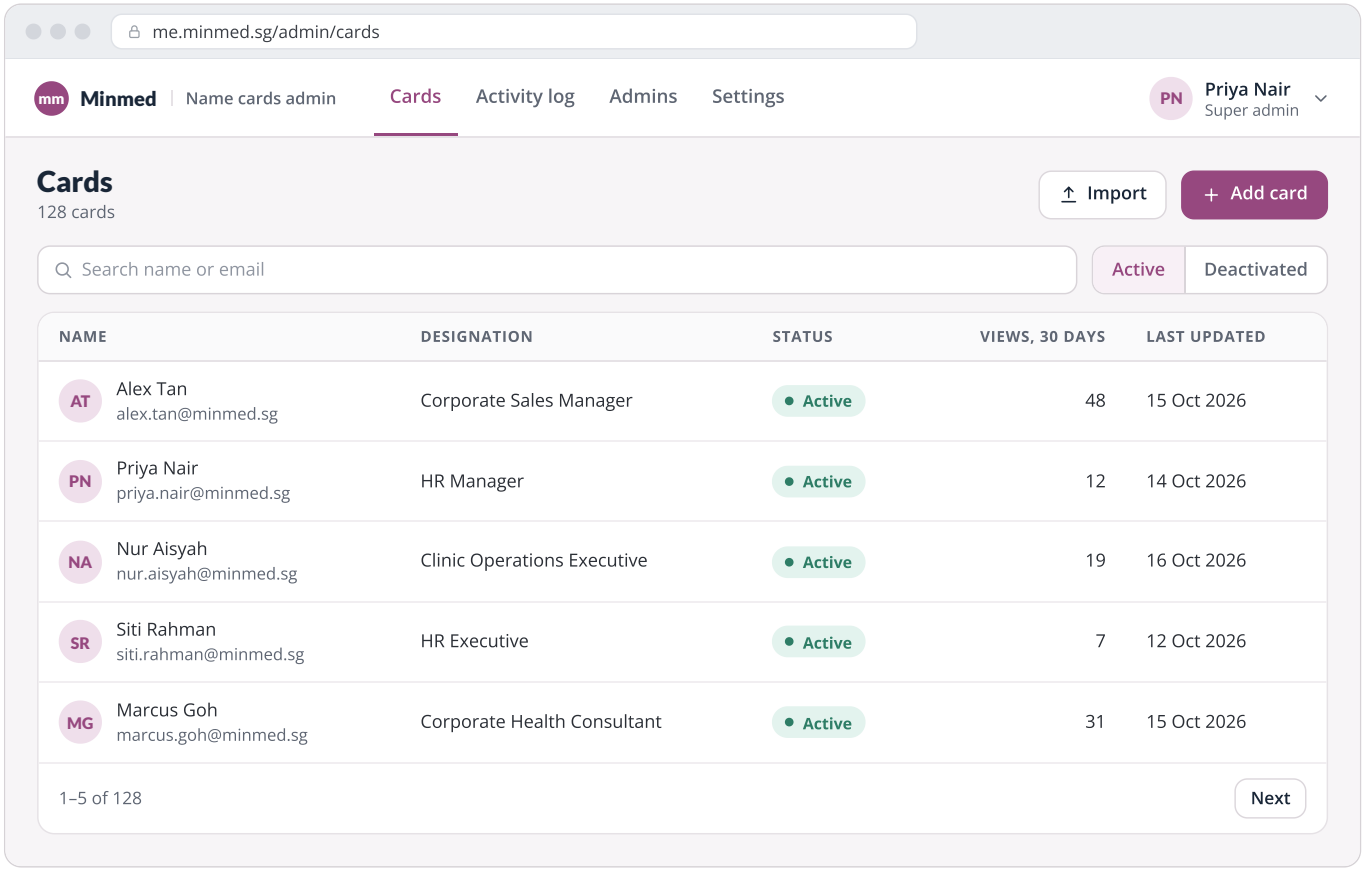


Figure 20. Cards.
`/admin/cards`. Search by name or email. Deactivated cards are soft-deleted: they leave this list and appear under Deactivated, for Super admins. Views count card page openings in the last 30 days.

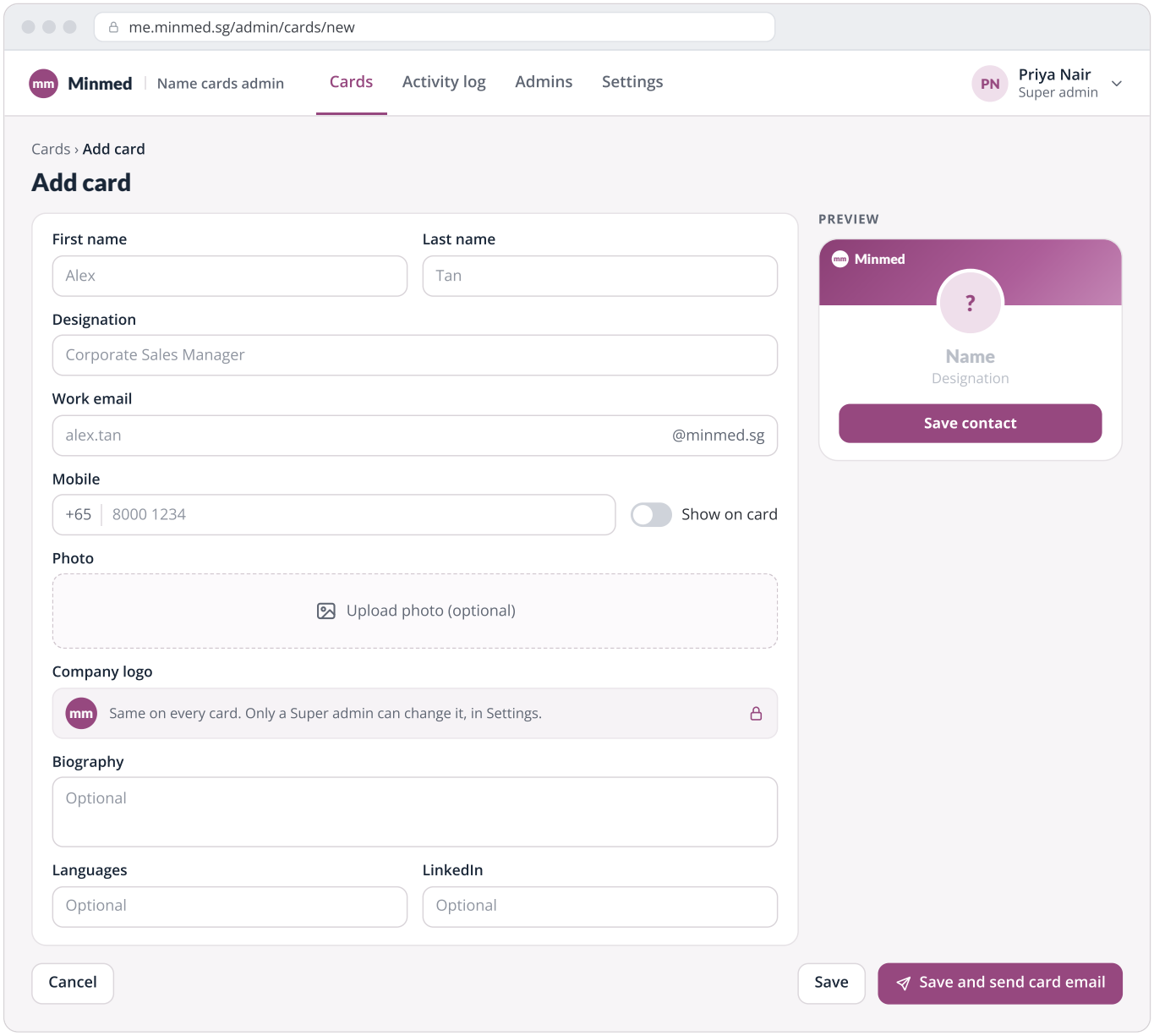


Figure 21. Add card.
`/admin/cards/new`. Work email must end with @minmed.sg. The mobile shows on the card only when switched on. The company logo is locked: it's the same on every card.

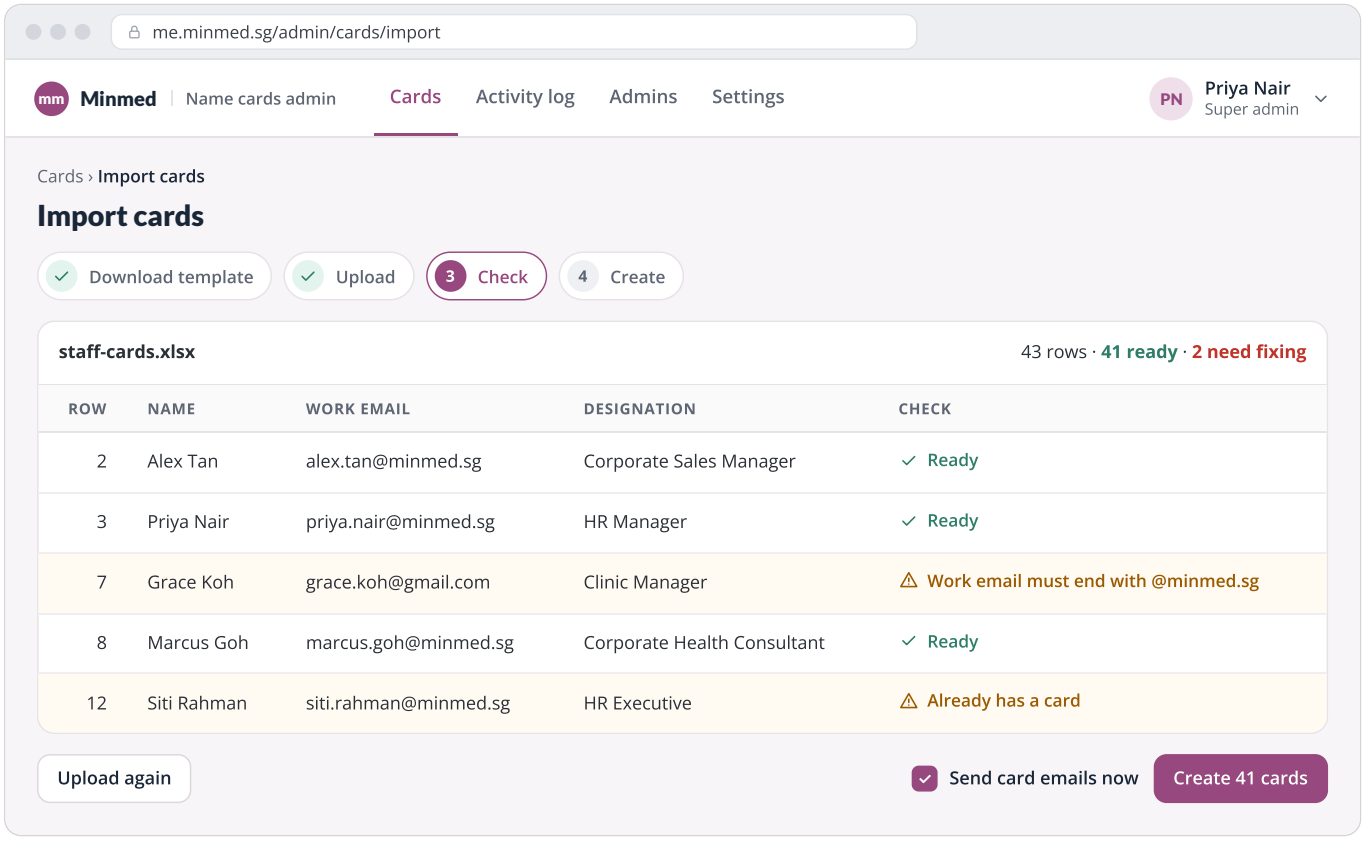


Figure 22. Import cards.
`/admin/cards/import`. Rows with problems are skipped until fixed. The card emails can go out in the same step.

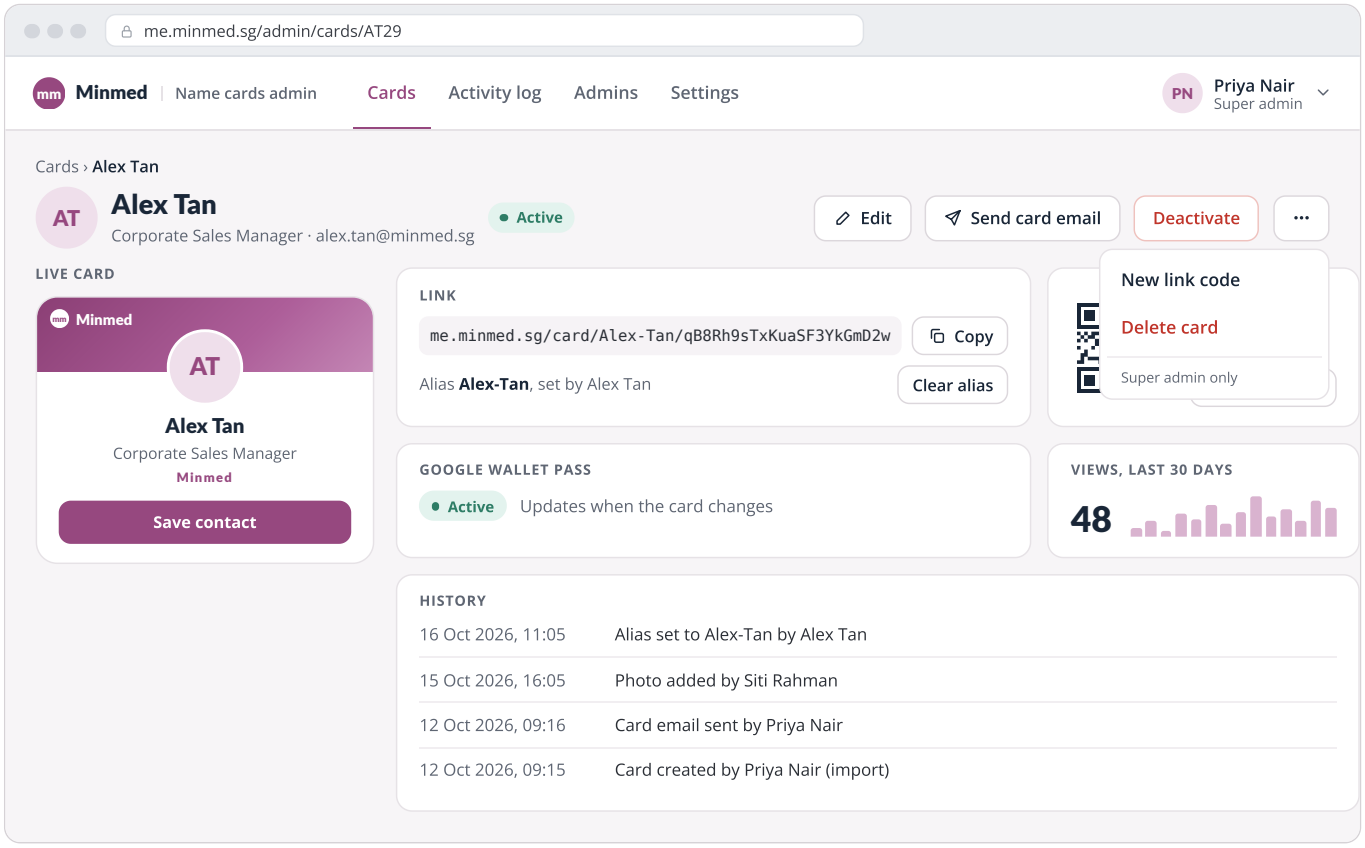


Figure 23. Card details.
`/admin/cards/{id}`. Admins can clear an alias. The ⋯ menu (New link code, Delete card) appears only for Super admins. On a deactivated card, Restore replaces Deactivate.

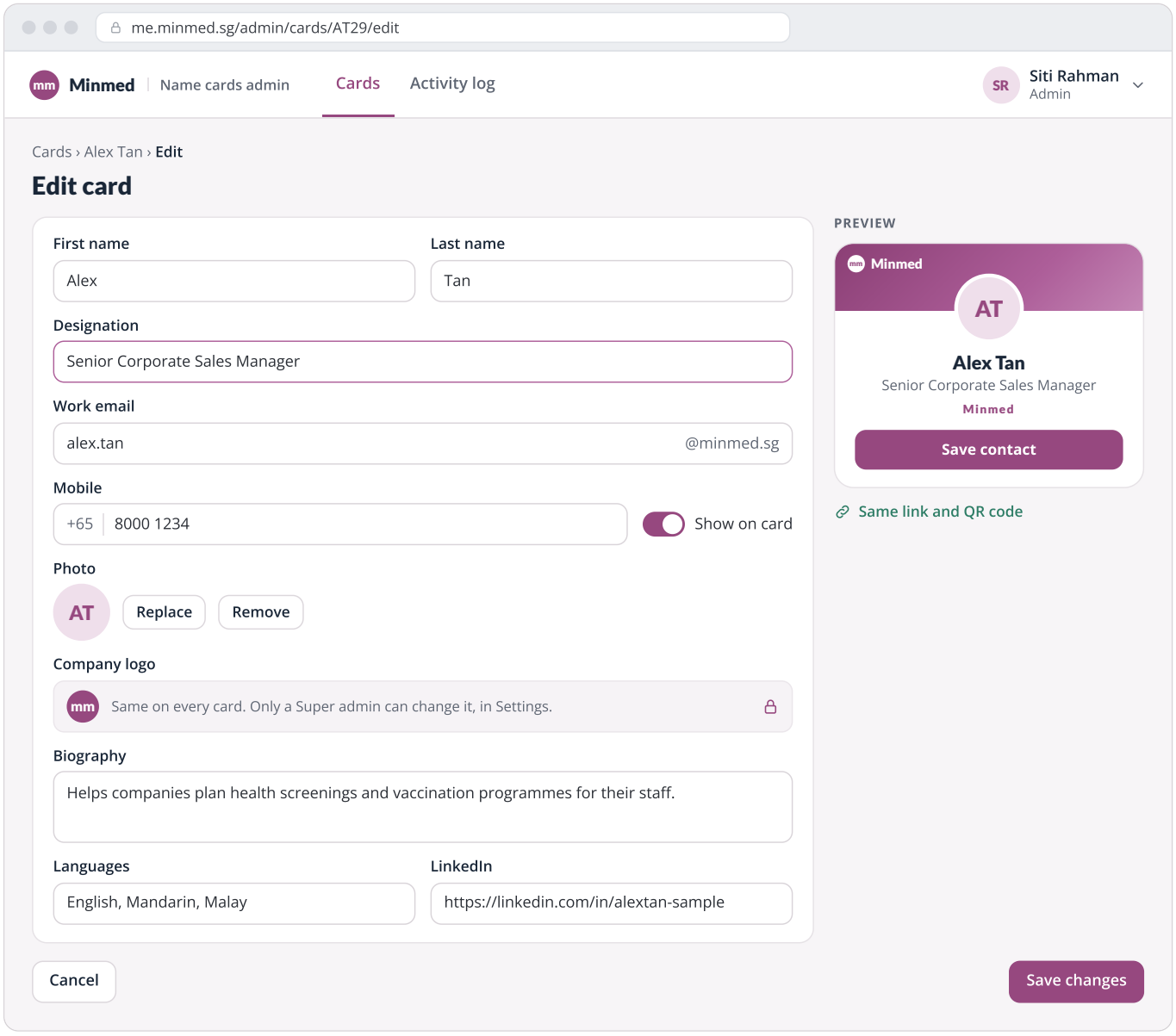


Figure 24. Edit card.
`/admin/cards/{id}/edit`, seen by an Admin, who has no Settings tab. Saving changes the live card at once, on the same link and QR code, and emails manage@minmed.sg. The company logo is locked here for everyone.

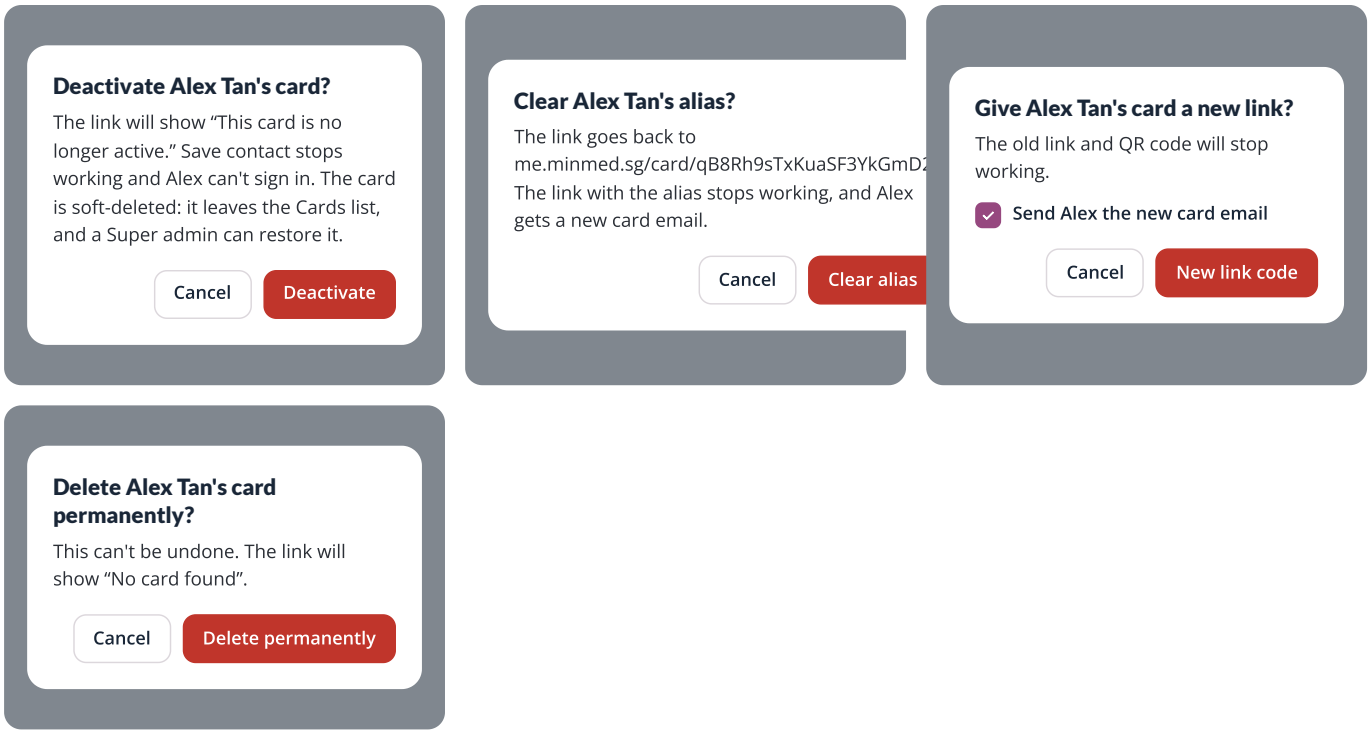


Figure 25. Confirm dialogs.
Deactivate and Clear alias are for Admins and Super admins. New link code and Delete are for Super admins only.

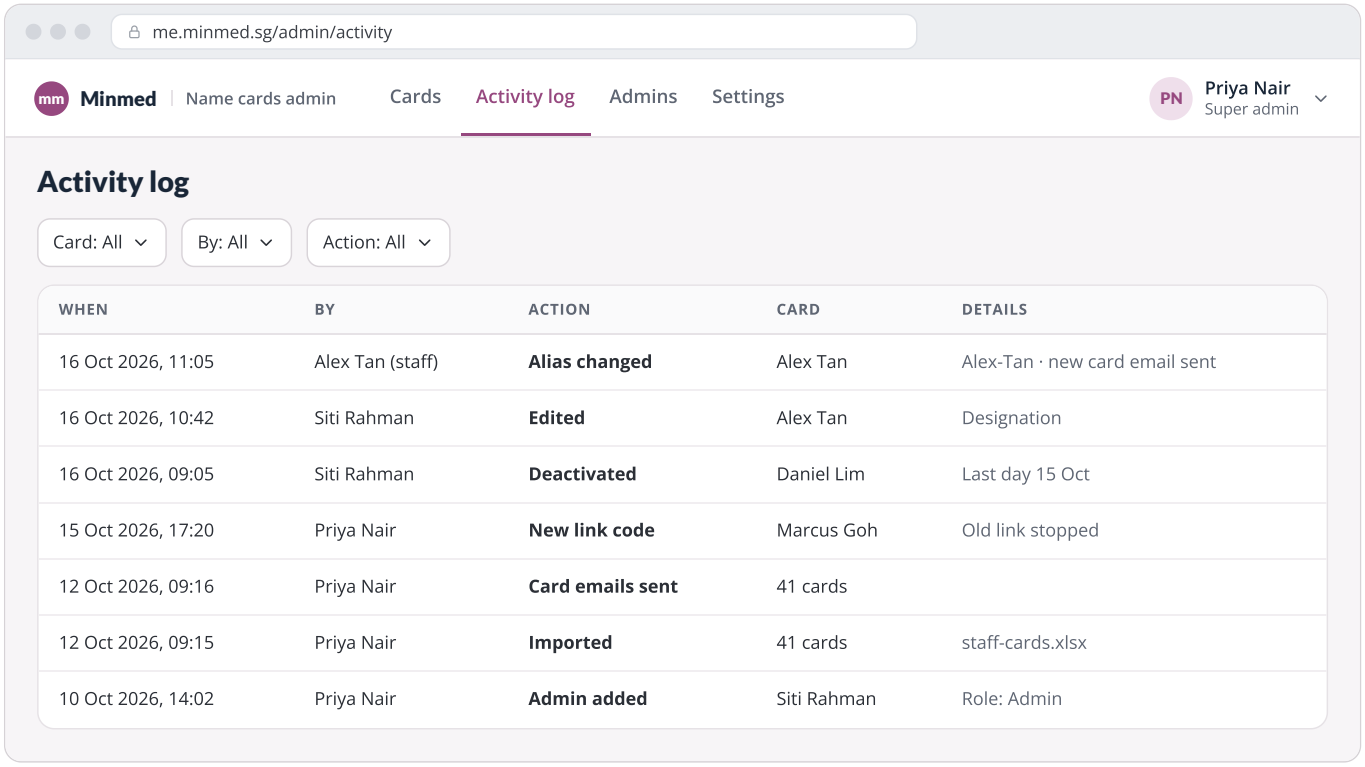


Figure 26. Activity log.
`/admin/activity`. Every admin action and alias change: who, what, which card and when.

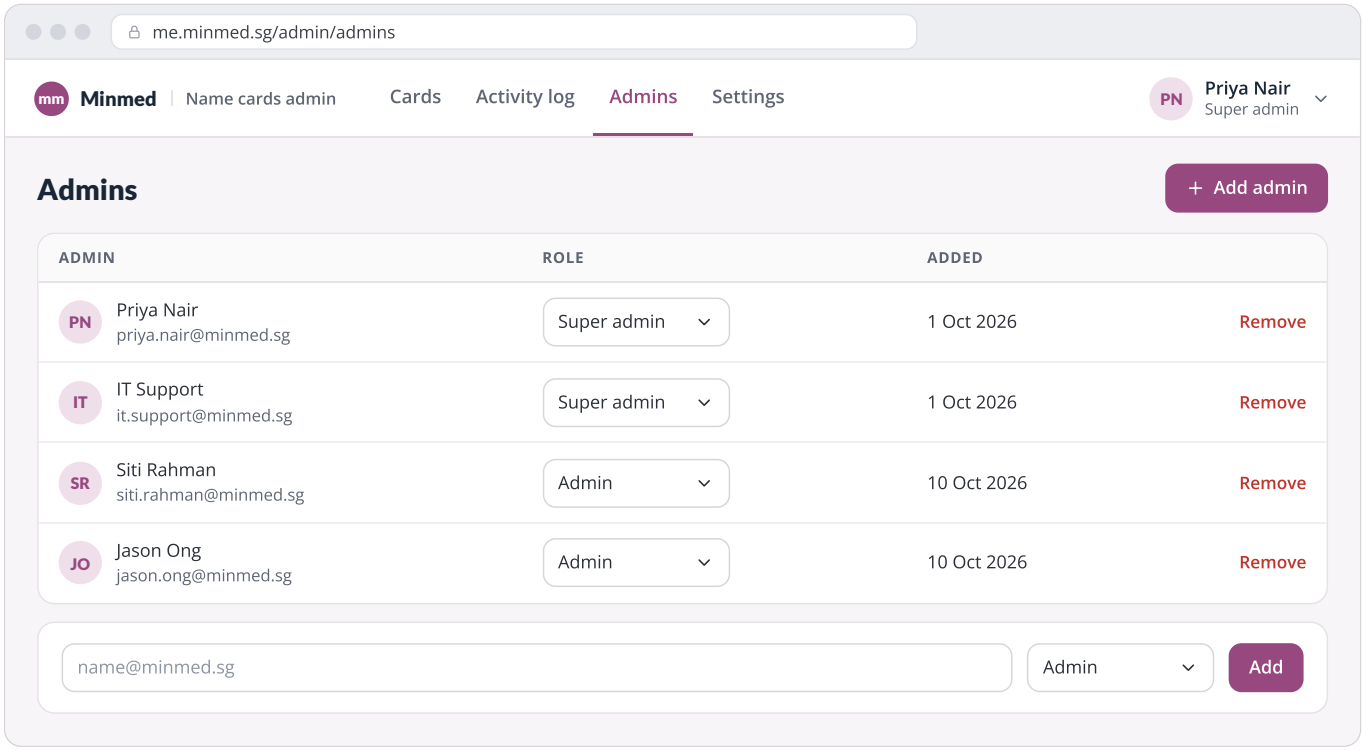


Figure 27. Admins (Super admin only).
`/admin/admins`. Admins are added by company email.

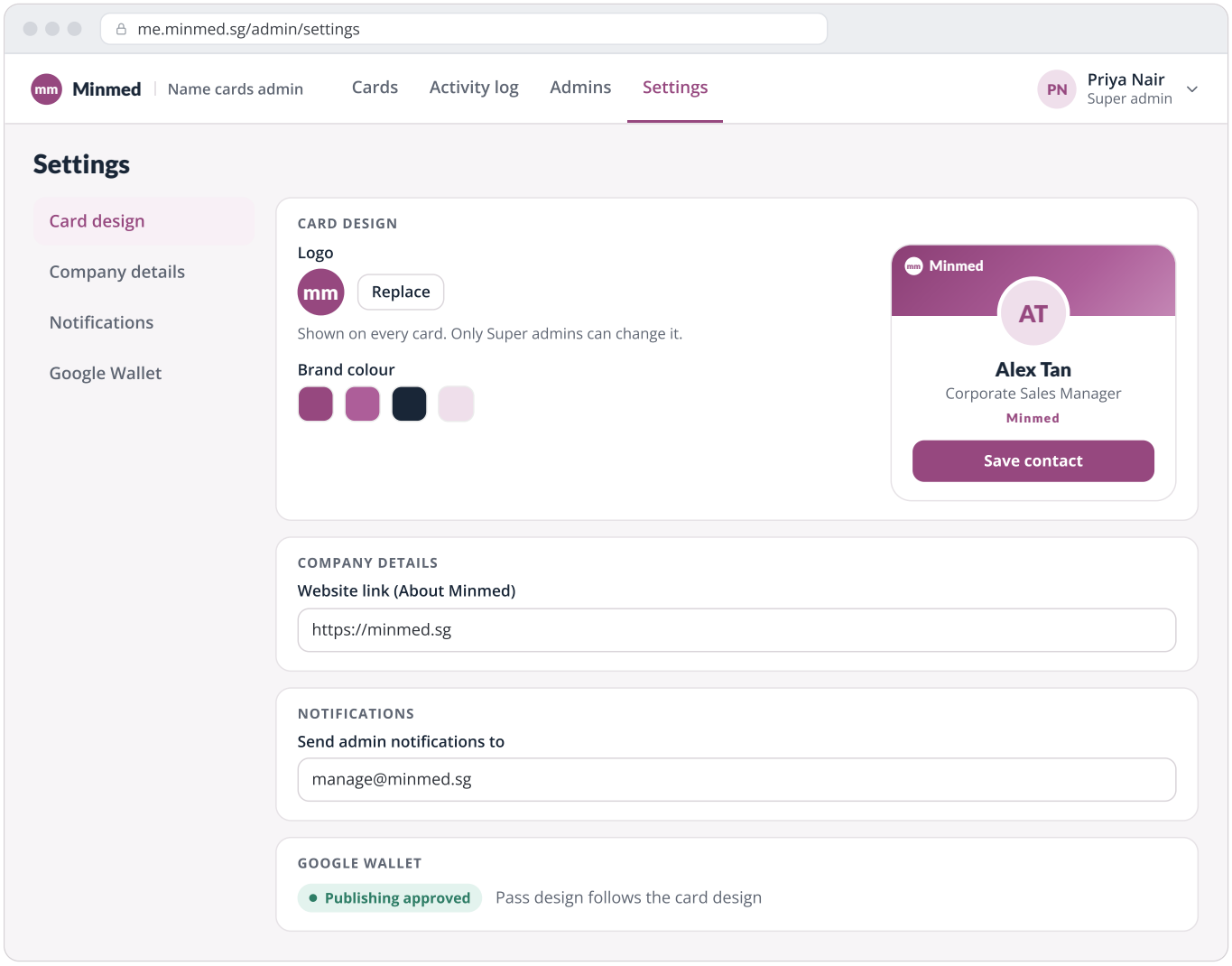


Figure 28. Settings (Super admin only).
`/admin/settings`. Card design: the company logo on every card, which only Super admins can change. Also company details, notifications and Google Wallet.

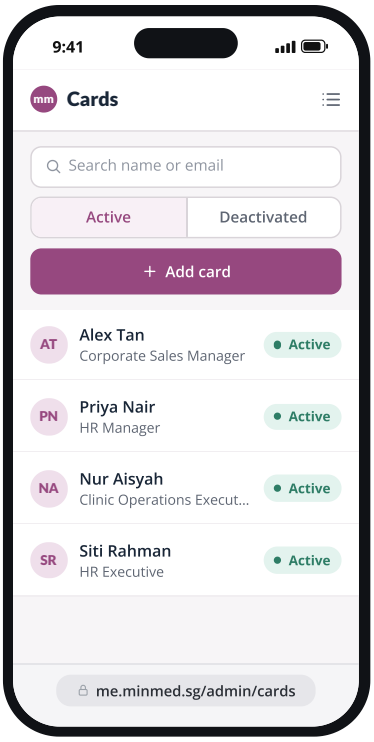


Figure 29. Admin console on a phone.
The same Cards page on a phone: the table becomes a list, with the same Active and Deactivated tabs.

Card edited: Alex Tan

From Minmed Name Cards **To** manage@minmed.sg

Card	Alex Tan
Change	Designation: ~~Corporate Sales Manager~~ → Senior Corporate Sales Manager
By	Siti Rahman (Admin)
When	16 Oct 2026, 10:42 am

Open in admin console

Alias changed: Alex Tan

From Minmed Name Cards **To** manage@minmed.sg

Card	Alex Tan
Change	Link: ~~me.minmed.sg/card/qB8Rh9sTxKuaSF3YkGmD2w~~ → me.minmed.sg/card/Alex-Tan/qB8Rh9sTxKuaSF3YkGmD2w
By	Alex Tan (staff)
When	16 Oct 2026, 11:05 am

Open in admin console

Figure 30. Admin notification.
Sent to manage@minmed.sg for every add, edit, alias change, deactivation, restore, new link code and deletion. Two examples: an admin's edit and a staff member's alias change.

8

Bot protection (CAPTCHA)

Purpose: slow down bots that copy cards in bulk. Random links and hiding from search stop people finding cards; these rules limit how fast anyone can open them. A CAPTCHA can't stop a person, or a bot that pays a solving service, from copying a card they can already open.

Rule	What's counted	Limit	Then	Status
Rule 1 **Same IP**	Card opens, contact downloads and wrong card addresses from one IP address	More than 3 in a rolling minute	CAPTCHA before anything is sent	CONFIRMED
Rule 2 **Pass**	A browser that has passed the CAPTCHA	30 minutes	No new CAPTCHA for that browser	CONFIRMED
Rule 3 **Busy card**	Opens and downloads of one card, from all IP addresses together	More than 20 in a rolling 10 minutes	Everyone opening that card gets the CAPTCHA for the next 30 minutes	CONFIRMED Replaces "every 5th visitor"
Rule 4 **Limit after a pass**	Opens and downloads by a browser holding a pass	More than 20 in 5 minutes	The pass ends and the CAPTCHA comes back	CONFIRMED
Never checked	WhatsApp and other link-preview fetches (they only see the preview), the admin console and My card (Google sign-in), and images, styles and scripts.			

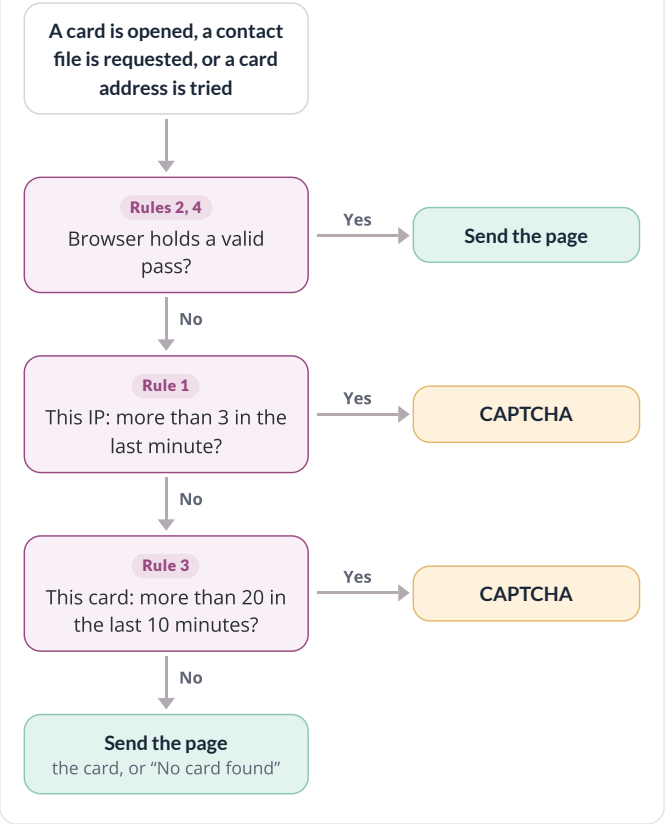


Figure 31. How each request is checked.
Counting happens on the server, in rolling windows.

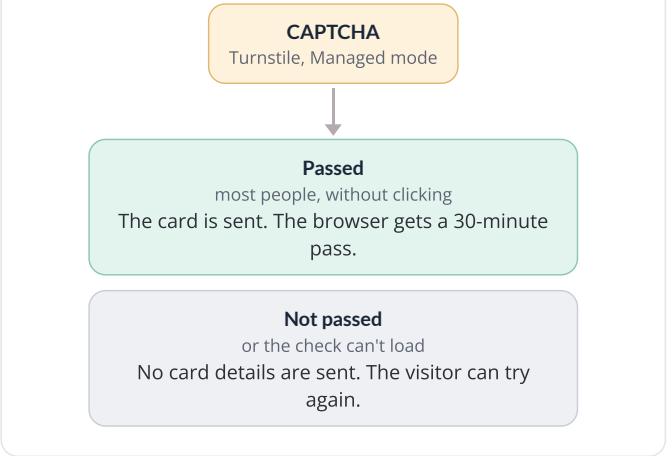


Figure 32. What happens at the CAPTCHA.
The page is in Figure 6. A Turnstile token is checked by a backend function; it works once and expires after 5 minutes.

Based on industry standards and research

Each rule follows published guidance from OWASP (a web security nonprofit), Cloudflare and Google.

Industry practice	Source	Used in
Use a CAPTCHA only as a step-up for suspicious traffic, not as the main defence. None of the sources suggests checking every Nth visitor: that would interrupt 1 in 5 real people, while a bot just tries again.	OWASP	Rules 1 and 3, instead of “every 5th visitor”
Limit by more than the IP address. For example, count all requests to one page, whoever sends them.	OWASP, Cloudflare	Rule 3
Count in rolling windows, not fixed ones, so bursts can't hide at the edges.	OWASP	All rules
A pass lasts 30 minutes by default; 15 to 45 minutes is recommended.	Cloudflare	Rule 2
Keep limits in place after a CAPTCHA is passed.	OWASP	Rule 4
Against scraping, use two limits per IP: 10 requests in 2 minutes, then 20 in 5 minutes.	Cloudflare	Rule 4's limit
Prefer managed checks that only ask a person to click when needed.	Cloudflare	Turnstile in Managed mode
Run the checks without acting first, then set the limits from real traffic.	Google, Cloudflare	The first 2 weeks record only

The numbers are starting points. Rule 4's limit matches Cloudflare's example; rule 3's is our estimate for a name card. Both get checked during the record-only weeks.

- **Managed mode:** Turnstile decides whether to show a checkbox. Most people never see it, and there are no picture puzzles.
- **Shared networks:** office Wi-Fi, events and mobile data put many people behind one IP address, so they meet rule 1 sooner. So does one person who opens a card, saves it and reloads twice. In Managed mode, most still pass without clicking.
- **Server-side:** the card page asks a backend function for the card. When a check is needed, no card details are sent until Cloudflare confirms the token.
- **Launch** CONFIRMED: for the first 2 weeks the rules only record. Compare them with real traffic, then switch them on.

Sources: OWASP bot management, Cloudflare rate limiting best practices, Cloudflare challenge passage, Cloudflare challenge types, Google reCAPTCHA v3, Turnstile widget modes and server-side validation.

9

Links and privacy

- **Link format:** `me.minmed.sg/card/{code}`, or `me.minmed.sg/card/{alias}/{code}` once the staff member adds a name or alias. For example `me.minmed.sg/card/qB8Rh9sTxKuaSF3YkGmD2w` and `me.minmed.sg/card/Alex-Tan/qB8Rh9sTxKuaSF3YkGmD2w`. The code has no name, initials or counting numbers. The `/card/` prefix keeps clear of Base44's own addresses.
- **The code** CONFIRMED: 22 random letters and numbers, made by the server with a secure random generator (CSPRNG) and never reused. That's about 131 bits of randomness, above the 128-bit minimum in the standard below.
- **The alias:** optional and chosen by the staff member: letters, numbers and hyphens, 2 to 30 characters. It adds no security; the code alone keeps the link unguessable. A link with an alias can be found by searching for that alias.
- **Both parts are checked:** the code must belong to a card, and the alias must be the card's current one, or absent when none is set. Anything else shows “No card found” and counts towards rule 1.
- **The link changes only** when the alias is set, changed or cleared, or a Super admin issues a new code. Name and designation edits keep the same link.
- **An alias change ends the old link:** links already sent, printed QR codes and Apple Wallet passes made from it show “No card found”. The Google Wallet pass updates itself and a new card email goes out.
- **Every public address uses the same random code:** the card page, contact file, card image and Google Wallet pass. Never a counting number, so nobody can fetch cards by counting.
- **Hidden from Google, twice over:** no list of cards anywhere (no sitemap entries, no staff directory, the home address redirects to minmed.sg), and every card page, image and contact file tells search engines not to index it. On Base44 that means the page's “Search visibility” switch plus a noindex header on files from backend functions, checked after every publish. Don't block card addresses in robots.txt: Google then can't see the “don't index” instruction and may still list a link someone posted elsewhere.
- **If a link is misused:** a Super admin issues a new code. The old link then shows “No card found”. An admin can also clear an offensive alias.
- **Bots:** see section 8.

Based on industry standards and research

Industry practice	Source	Used in
A link that is the only key to a page must be unguessable: a random identifier, never a counting number.	W3C	Random code
Random strings meant to be unguessable need a secure random generator and at least 128 bits of randomness. Each bit doubles the number of possible codes.	OWASP ASVS 5.0	22 characters
A fixed or predictable part, such as a name or initials, adds no strength.	OWASP	The alias adds none; the code alone keeps links unguessable
Researchers scanned every possible 5- and 6-character short-link code. They found shared OneDrive files and Google Maps directions that revealed people's home addresses.	Georgiev and Shmatikov, 2016	Long codes; rule 1 counts wrong addresses

Sources: W3C Good Practices for Capability URLs (2014 draft), OWASP ASVS 5.0 requirement 11.5.1, OWASP session management and Gone in Six Characters: Short URLs Considered Harmful for Cloud Services.

10

Feasibility on Base44 and Wallet

Checked against Base44, Apple, Google and Cloudflare documentation on 16/10/2026. Base44's settings are new and its docs contradict each other in places, so anything marked TEST needs trying on a throwaway app before the build starts. The WhatsApp preview options and the tests are in the build notes, a separate file.

Capability	Status	What it means for the scope
Custom domain me.minmed.sg	WORKS	One DNS record pointing to Base44, on the Starter plan or higher. If minmed.sg's DNS is on Cloudflare, the record must be “DNS only”, so Cloudflare's own protection can't sit in front.
Card address with optional alias	TEST	Base44 keeps addresses at the root for sign-in and system pages, so card links need a prefix: `/card/{code}` and `/card/{alias}/{code}`. Two-part addresses aren't documented.
Hidden from Google	TEST	A per-page “Search visibility” switch (added 28/09/2026) marks pages noindex and removes them from the sitemap. Files served by backend functions can carry their own noindex header. Check what the live site serves after every publish.
Per-card WhatsApp preview	NOT SUPPORTED AS IS	Base44 sets preview tags per page template, not per card, so every card would share one Minmed preview. The options and plan are in the build notes.
CAPTCHA (Cloudflare Turnstile)	TEST	Free plan: unlimited checks, up to 20 widgets, and no need to route traffic through Cloudflare. The page loads Turnstile (Cloudflare suggests a React component); a backend function verifies each token using a secret key kept in Base44's secrets. Base44's security policy can't be changed per app, so check that Turnstile loads.
Visitor IP address	TEST	Rule 1 needs the visitor's real IP address inside a backend function. Base44 passes the request headers on, but doesn't document which one holds the visitor's IP. Log the headers on a test app. If none does, rule 1 needs option C's separate host.
Counting for the CAPTCHA rules	OURS TO BUILD	Cloudflare can't sit in front of a Base44 domain, so the rules are counted in backend functions with a database table, in rolling windows.
Contact file and Google Wallet link	TEST	Backend functions can return files of the right type and use code libraries. Their addresses look like `me.minmed.sg/functions/contact?code=…`, still behind the random code.
Sign-in limited to minmed.sg (admins and staff)	WORKS, WITH SETUP	Needs Minmed's own Google sign-in set to “Internal” (Builder plan or higher). Base44 only has “admin” and “user” roles, so Super admin and Admin are our own list, editable only by Super admins. Staff sign in as ordinary users. Base44's plans don't charge per user.
Staff limited to their own card and alias	TEST	Base44's row-level rules can limit a signed-in user to the card with their email, and field-level rules can limit which fields they write. Saving goes through a backend function that checks the alias, then updates the link, the Google Wallet pass, the activity log and the emails. Check that the field rule can be tied to the person's own card.
Soft delete for leavers	WORKS	Base44 records carry a soft-delete flag and date, so a deactivated card can be hidden but kept, and restored by a Super admin.
Visitors can't list cards	OURS TO BUILD	A public Base44 app lets anyone register. Visitors must only get one card's public fields, by its code, through a backend function. One wrong rule could expose every card.
Emails to staff and manage@minmed.sg	WORKS, WITH SETUP	Needs a paid plan and a verified sending domain. One recipient per email, about 1–2 credits each.

Capability	Status	What it means for the scope
View counts	WORKS	Record card openings ourselves; Base44's own analytics keeps 60 days. A view is a card page opening, not a person. Link previews and admin previews don't count.
Rate limits and credits	BUILT IN	Base44 rate-limits public endpoints by default; the limits aren't published or adjustable. Function calls, counting and emails use credits. If credits run out, those actions pause and visitors see the temporary error page. Watch usage.
Google Wallet	TEST	Free issuer account, but Google must approve publishing (needs a business profile). Passes can be updated and set to expired.
Apple Wallet	TEST	iOS 27's Create a Pass: from a screenshot of the QR code on iPhone 15 Pro or newer with Apple Intelligence; on other iOS 27 iPhones (iPhone 11 or newer), Create Pass Manually and scan the QR code. No Apple fee. Minmed can't update or remove these passes.

Sources: Base44 docs on external domains, search visibility, SEO and sharing, backend functions, data security, sending emails, security, troubleshooting and credits; Apple's Create passes in Wallet; Google's Generic pass, publishing access and expired passes; Cloudflare's Turnstile plans and Turnstile overview.

11

Open items

The only open items. Everything else is confirmed.

1 **WhatsApp previews: option A, B or C.** TEST

Plan confirmed: test B first. Then C if per-card previews are a must, otherwise A. The options are described in the build notes. Affects Figure 10, section 10 and the file addresses in section 6. Needs the test result.

2 **Wording on the preview screen.** PROPOSED

Step 2 of Figure 17: the title Preview, the note “Not saved yet. This is how your card will look at your new link.”, the new link shown above the card, and Back and Save link buttons. The Preview button sits next to Save link (Figure 16, Figure 18). Needs your OK.

Sources. Minmed Digital Name Cards Technical Specification draft 0.2 (15/10/2026). Your decisions in this conversation on 16/10/2026. Standards, research, platform and vendor documentation linked in sections 8, 9 and 10.